

Program overview

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Year 2023/2024
 Organization Technology, Policy and Management
 Education Bachelor Systems Engineering, Policy Analysis & M

Code	Omschrijving	ECTS	p1 p2 p3 p4 p5
Bachelor TB 2023			
1e jaar TB 2023	1e jaar BSc TB 2023		
TB111C	Problem Analysis	5	
TB112C	Systems Modelling 1: Modelling Principles	5	
TB113B	Systems Modelling 2: Complex Systems	5	
TB121B	Governance and Law 1	5	
TB122B	Micro and Market Economy	5	
TB131B	Differential Equations and Linear Algebra	5	
TB132B	Multivariable Analysis and Linear Algebra	5	
TB133D	Introduction to Programming with Python	5	
TB134A	Statistics and Data Analysis	5	
TB135C	Operational Research	5	
Technologiespecialisatie 1	Technology Specialisation 1		
TB141EB	Introduction to Energy and Industry Systems	5	
TB141ID	ICT System Engineering and Rapid Prototyping	5	
TB141TB	Transport System Analysis	5	
Technologiespecialisatie 2	Technology Specialisation 2		
TB142EA	Analysis of Energy Systems	5	
TB142IC	Data Science in Technische Bestuurskunde	5	
TB142TA	Logistics 1	5	
2e jaar BSc TB 2023	2nd year BSc TB 2023		
TB211A	Analysis of Multi-Actor Systems	5	
TB212B	Ethics and Safety	5	
TB221B	Economy of Infrastructures	5	
TB222A	Governance and Law 2	5	
TB223D	Organisation and Management	5	
TB231C	System modelling 3: simulation methods	5	
TB232B	Research and Data-analysis	5	
TB233B	System modelling 4: project simulation methods	5	
TB234B	Multivariate Data Analysis	5	
Technologiespecialisatie 3	Technology Specialisation 3		
TB241EA	Transport Phenomena	5	
TB241IB	Object Oriented Programming	5	
TB241TB	Logistics 2	5	
Technologiespecialisatie 4	Technology Specialisation 4		
TB242EB	Processes in the Energy Sector	5	
TB242IC	Interconnected Engineering Systems	5	
TB242TB	Improving the Transport System	5	
Technologiespecialisatie 5	Technology Specialisation 5		
CTB1420-17	Transport en Planning	5	
TB243EA	Analysis of Industrial Systems	5	
TB243IB	I&C Systems: Risk & Control	5	
3e jaar BSc TB 2023	3rd year BSc TB 2023		
Verplichte vakken 3e jaar 2023			
TB323A	Governance van sociotechnische systemen	5	
TB351D	Bachelor Final Project	15	
Governance specialisatie (E&I / I&C / T&L)	Governance specialisation (E&I / I&C / T&L)		
TB321EB	Governance specialisation (E and I)	5	
TB321IB	Governance specialisation (I and C)	5	
TB321TB	Governance specialisation (T and L)	5	
Technologiespecialisatie 6	Technology Specialisation 6		
TB341EC	Performance analysis in Energy and Industry	5	
TB341ID	I&C in a socio-technical context	5	
TB341TC	Quantitative Models for Transport	5	

Year	2023/2024
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Bachelor TB 2023

Program Title	Technische Bestuurskunde
Director of Education	A.M.G. (Anneke) Zuiderwijk-van Eijk
Prerequisites	
Introduction 1	https://www.tudelft.nl/studenten/faculteiten/tbm-studentenportal/onderwijs/bachelor/welkom-nieuwe-bachelor-studenten

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1e jaar TB 2023	
Director of Studies	A.M.G. Zuiderwijk-van Eijk

TB111C	Problem Analysis	5
Module Manager	Dr. L.J. Kortmann	
Instructor	M.B. van der Meulen	
Instructor	N. van der Linden	
Co-responsible for assignments	N. van der Linden	
Contact Hours / Week x/x/x/x	5/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	

TB112C	Systems Modelling 1: Modelling Principles	5
Module Manager	Dr. P.W.G. Bots	
Instructor	Dr. P.W.G. Bots	
Instructor	Dr.ir. I. Bouwmans	
Responsible for assignments	Dr. P.W.G. Bots	
Co-responsible for assignments	Dr.ir. I. Bouwmans	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	

TB113B	Systems Modelling 2: Complex Systems	5
Module Manager	Prof.dr. M.E. Warnier	
Instructor	Ö. Okur	
Responsible for assignments	Prof.dr. M.E. Warnier	
Co-responsible for assignments	Ö. Okur	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Course Contents	Dit vak verzorgt een introductie in het modelleren en simuleren van complexe socio-technische systemen.	
Study Goals	De volgende onderwerpen komen aan bod in het vak: complexiteit en socio-technische systemen, fysieke modellen vergeleken met socio-technische modellen, procesmodellen, simulatiemodellen, de modelleer cyclus, het concept agent en agent interactie, emergentie, wachtrijen, stockflows. Na succesvolle afronding van de module: - kan de student uitleggen wat een complex systeem is en hoe dit zich verhoudt tot andere systemen - is de student in staat om de informatie flow van een complex systeem te modeleren en beschrijven mbv een proces diagram in IDEF0/BPMN - is de student in staat om de modelcyclus die nodig is voor het maken van een simulatiemodel te doorlopen (conceptualisatie, specificatie, implementatie, verificatie, validatie, experimenteren enz.); - kan de student van de belangrijkste modelleerconcepten (toestand, regels, invoer, uitvoer, visualisatie enz.) uitleggen hoe die gebruikt worden in basis simulatiemodellen, en kan de student python als simulatiegereedschap efficiënt en effectief gebruiken - is de student in staat een niet te complex model foutloos en binnen beperkte tijd in python te coderen - is de student in staat om voor een simulatie model een experiment plan op stellen dat systematisch het gedrag van het model exploreert, en dat gebruikt kan worden voor het beantwoorden van specifieke onderzoeksvragen	
Education Method	Colleges en practica.	
Assessment	Voor de practica wordt Python gebruikt. Het practica bereid voor op de stof van de modelleerlijn in het tweede jaar. Studenten werken in groepen van maximaal 5 studenten aan een drietal modelleeropdrachten. De modelleeropdrachten zijn in python en maken gebruik van de mesa bibliotheek. De eerste twee opdrachten zijn formatief, de laatste is summatief. Studenten moeten voor deze laatste opdracht minimaal een 5 halen en de opdracht telt voor 30% mee voor het eindcijfer van het vak. Tentamen (70%) en groepsopdracht (30%). Voor beiden onderdelen moet een score >5 worden gehaald. Daarnaast moet de skills module Information Literacy 2 met een voldoende zijn afgerond. Het vak is gehaald bij een eindcijfer > 5,75 Bij een onvoldoende voor de groepsopdracht is er een mogelijkheid om een nieuwe versie in te leveren. Deze nieuwe versie kan maximaal een 6 als eindcijfer krijgen. Indien er sprake is van onvoorziene omstandigheden, of maatregelen als gevolg van COVID-19, wordt als volgt afgeweken van voorgeschreven assessment: - het tentamen zal online worden afgenomen in de vorm van een open boek tentamen waar studenten drie uur de tijd voor hebben om het te maken. - de groepsopdracht blijft hetzelfde	

TB121B	Governance and Law 1	5
Module Manager	mr. J.M. Kooijman	
Instructor	Dr.ir. E. Minkman	
Responsible for assignments	mr. J.M. Kooijman	
Co-responsible for assignments	Dr. H.G. van der Voort	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Summary	The course's instruction language is Dutch, please view the Dutch version of the study guide.	

TB122B	Micro and Market Economy	5
Module Manager	S. Renes	
Instructor	Dr. A.F. Correlje	
Instructor	S. Renes	
Instructor	Dr.ir. R. van Bergem	
Responsible for assignments	S. Renes	
Co-responsible for assignments	Dr.ir. R. van Bergem	
Contact Hours / Week x/x/x/x	0/3/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	

TB131B	Differential Equations and Linear Algebra	5
Module Manager	Dr. T.W.C. Vroegrijk	
Responsible for assignments	Dr. T.W.C. Vroegrijk	
Co-responsible for assignments	Drs. J.A. Verheij	
Contact Hours / Week x/x/x/x	5/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	

TB132B	Multivariable Analysis and Linear Algebra	5
Module Manager	Dr. T.W.C. Vroegrijk	
Responsible for assignments	Dr. T.W.C. Vroegrijk	
Co-responsible for assignments	Drs. J.A. Verheij	
Contact Hours / Week x/x/x/x	0/5/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	

TB133D	Introduction to Programming with Python	5
Module Manager	Ö. Okur	
Instructor	Ö. Okur	
Instructor	Dr.ir. H. Torkamaan	
Responsible for assignments	Ö. Okur	
Co-responsible for assignments	Prof.dr. M.E. Warnier	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course provides an introduction to programming. The course covers the following parts: - Python and Jupyter Notebooks - Basic imperative programming concepts (expressions, statements, functions, conditionals, loops) - Basic data types (strings, lists, dictionaries, tuples, files) - Debugging - Object-oriented programming - Introduction to libraries (Numpys, Matplotlib) This course is taught entirely in English.	
Study Goals	By the end this course, you should be able to: - Implement Python programs using basic data types and programming concepts - Apply Python programs using classes and objects - Implement Python programs using algorithms - Perform debugging to remove errors from complex Python programs	
Education Method	Mix of theory lectures and supervised exercises sessions. Programming assignments will be given which are used to practice the programming skills and will be worked on during the exercise sessions and in assignments. Reading material will be available.	
Literature and Study Materials	Book: Think Python How to Think Like a Computer Scientist - Allen Downey (Available online: https://greenteapress.com/thinkpython2/thinkpython2.pdf) Jupyter Notebooks with teaching material and practice assignments.	
Assessment	Computer exam	

TB134A	Statistics and Data Analysis	5
Module Manager	M. Kroesen	
Instructor	Dr. T.W.C. Vroegrijk	
Responsible for assignments	M. Kroesen	
Co-responsible for assignments	Dr. T.W.C. Vroegrijk	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	

TB135C	Operational Research		5
Module Manager	Dr.ir. P.W. Heijnen		
Instructor	Dr. P.W.G. Bots		
Instructor	Dr.ir. P.W. Heijnen		
Responsible for assignments	Dr.ir. P.W. Heijnen		
Co-responsible for assignments	Dr. P.W.G. Bots		
Contact Hours / Week x/x/x/x	0/0/0/4,75		
Education Period	4		
Start Education	4		
Exam Period	4 5		
Course Language	Dutch		
Expected prior knowledge	--		
Parts	--		
Course Contents	--		
Study Goals	--		
Education Method	--		
Assessment	--		

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Technologiespecialisatie 1	
Director of Studies	A.M.G. Zuiderwijk-van Eijk

TB141EB	Introduction to Energy and Industry Systems	5
Module Manager	Dr.ir. E.J.L. Chappin	
Instructor	Prof.dr.ir. M.P.C. Weijnen	
Responsible for assignments	Dr.ir. E.J.L. Chappin	
Co-responsible for assignments	Prof.dr.ir. M.P.C. Weijnen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Course Contents	--	
Study Goals	--	
Education Method	--	
Literature and Study Materials	--	

TB141ID	ICT System Engineering and Rapid Prototyping	5
Module Manager	S. Azimi Rashti	
Instructor	Dr.ir. J. De Stefani	
Instructor	S. Azimi Rashti	
Co-responsible for assignments	Prof.dr.ir. M.F.W.H.A. Janssen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	The course introduces students to think and act as software engineer in TB context. The course integrated theory and practical work by developing a software prototype as final deliverable using the theory and practice behind software engineering, requirements formulating, and prototype assembling. Students are able to rapid prototype a functional software using low-code application. For this deliverable, the course presents and compare the most common software approaches, teaches how to develop diagrams using Unified Modelling Language (UML) diagrams, and, how to differentiate the most common programming paradigms. Finally the resulting prototype should be evaluated within the broader societal context.	
Study Goals	At the end of the course, the student should be able to: LO1: Describe the most common software design processes (at least Waterfall, Agile, Re-use based) LO2: Model real-life situations using UML diagrams (at least, use-cases, class and sequence diagrams) LO3: Explain the basic conceptual distinctions for characterizing IT systems, including the following: - o LO3.a: Categorize programming languages in different levels of programming approaches (at least procedural, object-oriented, functional) - o LO3.b: Define the basic components of computer hardware and hardware architectures LO4: Implement functional software using rapid prototyping approach and a low-code platform LO5: Identify the societal impact of the developed application.	
Education Method	Expected education methods: - Lectures (presenting slides) - UML practical work (making use-case, sequence and class diagrams) - Software Workshops in format of Questions and Answers (Q&A) - Low-code practical work (making a functional prototype) - Elevator pitch students presenting low-code prototyping	
Assessment	The course assessment is be composed of three parts: Summative assignment: 30% of the final grade, Exam: - Part I - UML Modeling: 30% of the final grade - Part II - System engineering: 40% of the final grade - Summative Assignment : Submission via Brightspace + Student presentations on campus - Exam: On campus Computer Exam (via Mobius) Passing grade: To pass the course, the average grade of the three parts (Summative assignments + Exam Part I + Exam Part II) needs to be 5.8 or higher. Moreover, each part needs to have a grade of 5 or higher.	

TB141TB	Transport System Analysis	5
Module Manager	B. Pudane	
Instructor	B. Pudane	
Instructor	Dr. S. Balakrishnan	
Co-responsible for assignments	J. Rezaei	
Contact Hours / Week x/x/x/x	0/0/5/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch English	
Course Contents	The objective of this course is to learn to understand and analyse transport systems. Transport systems have large positive impacts on our daily activities and quality of our lives as they ensure mobility, accessibility, and safety for people and companies and have enormous importance for the economy. At the same time, they are also associated with problems in the environment, traffic safety, health and congestion. It is essential to view the current transport system in the context of both its positive and negative impacts on environment, accessibility, health, and (traffic) safety. Policy makers and researchers often ask: how can we increase the accessibility that the transport system provides without exacerbating the problems of the system? Better still: how can we improve accessibility and at the same time solve congestion, environmental, safety, and health problems? In this course you will learn to analyse challenging policy questions such as the above. You will learn about the components of the transport system: technology (roads, cars, etc.), the behaviour of people and companies, and spatial developments. You will learn how these elements depend on each other, and how to conceptually analyse the societal impacts of interventions in the transport system (such as new policies or technological innovations). Finally, you will learn computation techniques and models that you can use to quantitatively estimate the impacts of any transport policy on environment (in particular, air quality), traffic safety, accessibility, and public health.	
Study Goals	Upon the successful completion of this course you should be able to: 1. describe the components of the transport system (such as travel behaviour, land use, individual needs) and their relationships, 2. discuss how transport activities relate to four key societal impact areas - environment, accessibility, safety, health & well-being, 3. appraise the societal impacts (on environment, accessibility, safety, health and well-being) of transport policies and technological scenarios via a customised conceptual model, 4. quantify the societal impacts of transport policies and technological scenarios following provided calculation models.	
Education Method	Students independently study the textbook "The Transport System and Transport Policy" (van Wee et al., 2023), as well as additional materials and videos provided by the teachers. During lectures, the students discuss and practice with the textbook material and work in groups on quantitative exercises. Groups are also expected to meet outside of the class to create a conceptual model for a specific intervention in the transport system, prepare a presentation and written assignment, and to complete the quantitative exercises. The group work is designed for the students to learn to analyse the transport system elements and their interrelationships, and to estimate effects quantitatively based on those relationships. Each group works independently while receiving support from the course instructors.	
Assessment	Learning objectives 1 to 4 are tested with an individual exam (60% of the final grade) and learning objectives 2 to 4 also with a group assignment (40%). In order to successfully complete this course, the weighted average grade must be at least 5.75, as well as the two components (exam and group assignment) must each be at least 5.0.	

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Technologiespecialisatie 2	
Director of Studies	A.M.G. Zuiderwijk-van Eijk

TB142EA	Analysis of Energy Systems	5
Module Manager	Dr.ir. L. Stougie	
Responsible for assignments	Dr.ir. L. Stougie	
Co-responsible for assignments	Dr.ir. E.J.L. Chappin	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	

TB142IC	Data Science in Technische Bestuurskunde	5
Module Manager	R.I.J. Dobbe	
Instructor	R.I.J. Dobbe	
Co-responsible for assignments	Dr.ir. G.A. de Reuver	
Contact Hours / Week x/x/x/x	0/0/0/14 - 6 in class, 8 homework	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	Dutch	
Course Contents	Today's information, communication and computing technology makes it possible for organizations to monitor their activities at almost every level, with the result that large amounts of data are collected, which in turn can form an input for other processes and applications. Processing collected data into information that is usable, reliable and secure is a delicate process. As a technical public administrator you must be able to make the translation between the administrative process and the technically available tools and methods. It requires an integrated look at the context of application as well as the life cycle of data, from collection to processing in analyzes and AI systems. In addition, it is essential to understand how AI systems behave and process data, and what ethical, legal and social implications these can have.	
Study Goals	LO1: Sociotechnical specification - Identify how AI systems may contribute to the formulation and solution of multi-stakeholder problems, by: a) determining and operationalizing a question such that an intelligent system or data analysis could produce a meaningful knowledge. b) exploring a data set to determine whether and how it might inform questions of interest or aid decision-making c) presenting the affordances and limitations of an AI system with appropriate visuals and written argumentation d) providing recommendations for how the AI system may be used in practice LO2: Design - Execute the design of an AI system for decision-making, including: a) selecting the appropriate modeling approaches and algorithms b) determining the need and quality of data inputs c) processing data sets using modern data science scripts and statistical tools d) building an AI model (rule-based or learning-based) for decision-making e) summarizing key mathematical concepts for AI systems f) apply relevant tests to assess a decision-making system for various metrics g) identifying the limitations of the AI system outputs and assessing its impact on correctness and utility LO3: Reflection - Communicating the affordances and limitations of a data-driven and algorithmic decision-making system in a responsible manner, including: a) using best practices for documenting the analyses for reproducibility and accountability b) reflecting on the ethical, legal and societal implications of an AI system in terms of the research process and implementation c) translating implications to possible tradeoffs in the problem formulation and use of the system	
Education Method	Main activities: - In-class instruction and discussion - Practical assignments - Group project - Reflection assignments	
Assessment	Practical lab assignments - 2/10 of grade A final reflection essay - 2/10 of grade A group project - 3/10 of grade A final exam - 3/10 of grade	

TB142TA	Logistics 1	5
Module Manager	Dr. J.H.R. van Duin	
Responsible for assignments	Dr. J.H.R. van Duin	
Co-responsible for assignments	I. Dursun	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Remarks	The lectures are partly taught in English and Dutch due to the composition of the teachers' team.	

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2e jaar BSc TB 2023	
Director of Studies	A.M.G. Zuiderwijk-van Eijk

TB211A	Analysis of Multi-Actor Systems	5
Module Manager	Dr.ir. V.J. Cortes Arevalo	
Instructor	Dr.ir. M.P.M. Ruijgh-van der Ploeg	
Instructor	Dr. L.J. Kortmann	
Instructor	Dr.ir. V.J. Cortes Arevalo	
Instructor	N. van der Linden	
Co-responsible for assignments	Dr. P.W.G. Bots	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch English	
Tags	Group work Intensive Policy Analysis Project	

TB212B	Ethics and Safety	5
Module Manager	Dr. S.M. Copeland	
Instructor	K.L.L. van Nunen	
Co-responsible for assignments	Prof.dr.ir. P.H.A.J.M. van Gelder	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch English	

TB221B	Economy of Infrastructures	5
Module Manager	Dr. A.F. Correlje	
Module Manager	Dr.ir. R. van Bergem	
Instructor	Dr.ir. R. van Bergem	
Instructor	S. Renes	
Responsible for assignments	Dr. A.F. Correlje	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	

TB222A	Governance and Law 2	5
Module Manager	mr. J.M. Kooijman	
Responsible for assignments	mr. J.M. Kooijman	
Co-responsible for assignments	Dr. H.G. van der Voort	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	

TB223D		Organisation and Management	5
Module Manager	Dr. M.L.C. de Bruijne		
Instructor	Prof.mr.dr. J.A. de Bruijn		
Instructor	Dr.ir. G. Bekebrede		
Responsible for assignments	Dr. M.L.C. de Bruijne		
Co-responsible for assignments	Dr.ir. G. Bekebrede		
Contact Hours / Week x/x/x/x	0/0/4/0		
Education Period	3		
Start Education	3		
Exam Period	3 4		
Course Language	Dutch		

TB231C	System modelling 3: simulation methods	5
Module Manager	Dr.ir. W.L. Auping	
Instructor	Dr.ir. I. Nikolic	
Instructor	Dr.ir. W.L. Auping	
Instructor	Dr. O. Kammouh	
Responsible for assignments	Dr.ir. W.L. Auping	
Co-responsible for assignments	F.M. d'Hont	
Co-responsible for assignments	Dr. M.D. - Kubli	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Required for	This course is required for students in the Technische bestuurskunde bachelor programme, and is a prerequisite for the project in TB233B Systems Modelling 4 simulation methods project.	
Expected prior knowledge	It is expected from students that they have basic knowledge about systems modelling, differential equations and linear algebra, and basic programming skills in Python.	
Parts	This course consists of two parts: - Agent-Based Modelling - System Dynamics	
Course Contents	In System Modelling 3: Simulation methods two different simulation methods will be taught: Agent-Based Modelling (ABM), and System Dynamics (SD). There will be a lot of emphasis on practising conceptualisation, formulation, evaluation, and use of models, in which the theory is aimed at head-starting students, and to better understand different concepts and background of these two modelling methods.	
Study Goals	Upon completion of this course, students should be able to - explain the process of modelling for supporting decision-making on large societal challenges; - conceptualise, formulate, parameterise, test, and use a model of a complex, societal policy problem using SD and ABM formalisms; - analyse a complex societal policy problem using an SD model and ABM; - identify and test interventions for addressing the policy problem using the developed simulation model; - assess and argue whether SD and/or ABM are appropriate for analysing a policy problem; - link system structure to possible dynamic behaviour of important indicators of a policy problem. For SD, this system structured is aggregated. For ABM, this system structure is microscopic. SD specific - develop and use a medium complexity SD model in Vensim ABM specific - develop and use a medium complexity ABM model in MESA.	
Education Method	This course uses two main methods of education: - Lectures for the theoretical aspect of the course - Lab sessions to develop skills for working with the tools Vensim and MESA	
Computer Use	Agent-Based Modelling: The ABM simulation models will be made with MESA. MESA is a Python 3 based framework for ABM. Student will need to have a working Python 3 installment before they install MESA. System Dynamics: The SD simulation models will be made with Vensim. Students can request a temporary Vensim license for Windows or iOS. This program is also installed in the computer rooms in the Faculty (TPM in Delft) and can be used via TU Delft's weblogin.	
Course Relations	This course expands on the knowledge obtained in TB131B Differential Equations and Linear Algebra, TB112C Systems modelling 1: modelleerprincipes, TB133D Introduction to Programming with Python, and TB113B Systems Modelling 2: Complex Systems. Especially the Python skills obtained during Systems Modelling 2 are indispensable during this course.	
Literature and Study Materials	Agent-Based Modelling: Jupyter notebooks and lecture handouts System Dynamics: E-book for theory and exercises	
Assessment	This course is examined in a written, online, open-book computer exam. The exam will be on campus. with two separate parts (with a break in between): - Agent-Based Modelling (50%) - System Dynamics (50%) The final grade is composed of partial grades for each part (weights as indicated on Osiris). Partial grades count when they are at least a 5.0 (compliant with the Rules and Guidelines of the Board of Examiners of the TB bachelor programme).	
Targetgroup	Students from the Technische Bestuurskunde bachelor programme.	

TB232B	Research and Data-analysis	5
Module Manager	M. Kroesen	
Instructor	Dr. E.J.E. Molin	
Instructor	Ir. S. van Cranenburgh	
Responsible for assignments	M. Kroesen	
Co-responsible for assignments	Dr. E.J.E. Molin	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	

TB233B	System modelling 4: project simulation methods	5
Module Manager	Ö. Okur	
Instructor	Dr.ir. I. Nikolic	
Instructor	Prof.dr. M.E. Warnier	
Instructor	Dr.ir. W.L. Auping	
Instructor	Ö. Okur	
Instructor	M.B. van der Meulen	
Instructor	Dr. O. Kammouh	
Co-responsible for assignments	Dr.ir. W.L. Auping	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	The prior knowledge for this course is developed during the following courses: - TB112C System's modelling 1: modelling principles; - TB113B System's modelling 2: complex systems; - TB231C System's modelling 3: simulation methods. Het last course, System's modelling 3, is the most important of these 3 courses.	
Course Contents	This project course will be primarily executed by groups containing 2 students. Each group will choose a modelling method: either system dynamics (SD), or agent-based modelling (ABM), and a socio-technical problem that they will model quantitatively. The socio-technical problem that will be selected will be designed such that it can be modelled both with SD and ABM. Each group will independently, but under the supervision of teaching assistants and the course lecturers, apply the different phases of the modelling cycle. Next, each group will use their own model to analyse a research problem and report this in a scientific article. In parallel to the modelling and simulation process, the academic writing skills of the students will be developed by the use of writing exercises and workshops. The skill level in academic writing will be assessed explicitly. Each group will receive the task to write peer reviews for a group that used the other modelling paradigm. In week 6, this will concern the model and the model description. In week 8, the results of both different methods can be compared, and the scientific article of the other group will be assessed on contents and writing skills.	
Assessment	The grade will be composed in the following way: - 20% academic writing - 20% model quality - 60% reporting quality - The two peer-reviews will count for the final grade and can, depending on their quality, result in a bonus or subtraction of the final grade within the range: [-0.5, +0.5]	

TB234B	Multivariate Data Analysis	5
Module Manager	Dr. E.J.E. Molin	
Responsible for assignments	Dr. E.J.E. Molin	
Co-responsible for assignments	M. Kroesen	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	

Year	2023/2024
Organization	Technology, Policy and Management
Education	Bachelor Systems Engineering, Policy Analysis & M

Technologiespecialisatie 3	
Director of Studies	A.M.G. Zuiderwijk-van Eijk

TB241EA	Transport Phenomena	5
Module Manager	Dr.ir. I. Bouwmans	
Instructor	Dr.ir. G. Korevaar	
Responsible for assignments	Dr.ir. I. Bouwmans	
Co-responsible for assignments	Dr.ir. G. Korevaar	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	

TB241IB	Object Oriented Programming	5
Module Manager	S. Azimi Rashti	
Instructor	S.H. van Engelenburg	
Instructor	Dr. N. Metoui	
Instructor	Dr.ir. J. De Stefani	
Instructor	S. Azimi Rashti	
Co-responsible for assignments	Dr.ir. G.A. de Reuver	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	This course requires: - Basic knowledge of UML Diagrams. - Basic knowledge of any (scripting or programming) language.	
Course Contents	This course deepens your skills and conceptual understanding of programming. You will learn object- oriented Design (OOD) with UML and Object-Oriented Programming (OOP) with Java. UML and Java are industry standard design and programming languages used to develop enterprise software, desktop apps, web apps, mobile apps, games and much more. In this course, you will learn how to position the languages you learned so far, and what to use when: Low-code (TB141Ic), scripting (TB133d; TB142Ic) and object-oriented (TB241Ib). You will also familiarize yourself with Java development tools and practices that allow you to write clean and reusable code and to understand and modify existing ones easily. The skill set you will earn in this course can be useful, in the future, to design and build software tools and components, to update or upscale software, or to manage a java project.	
Study Goals	At the end of this course, you will be able to: LO1 - Identify key characteristics of the Java programming language and development environment (EDI, SDK, JVM, etc.) LO2 - Use basic Java APIs and components (data types, data structures, algorithmic structures, and methods) LO3 - Apply key elements, concepts, and design patterns of object-oriented programming LO4 Write clean, understandable, and reusable code following Java coding best practices LO5 - Develop object-oriented programs using Java.	
Education Method	In this course, you will have alternating lectures and hands-on labs	
Literature and Study Materials	Materials (slides, book chapters, tutorials, references to software packages, etc.) will be made available via brightspace.	
Assessment	70% Individual tests and assignments The course has three paper based tests, where test 1 has 5%, test 2 10% and test 3 15% of the final grade of the course. There are also three coding assignments, where assignment 1 has 10%, assignment 2 15% and assignment 3 15% of the final grade of the course. 30% Group project The project will be carried out by (self-formed) groups of 5 students and will count for 30% of the final grade. The final project will have two deliverables: Code in a zip file Task repartition and progress report The students need at least 5 (15/30) to pass this part of the course. If the student get 15/30, s/he can keep the grade for next year if needed. If the student can less than 15/30, s/he has to retake the project next year. To pass the course, the final grade has to be 5.75 or higher. All partial grades have to be 5.0 or higher. Resubmission is not allowed for the individual and group assignments. In case of unforeseen circumstances or measures resulting from COVID-19, tests and assignments will be done (timed) online.	

TB241TB	Logistics 2	5
Module Manager	Dr. S. Fazi	
Co-responsible for assignments	I. Dursun	
Contact Hours / Week x/x/x/x	4/0/0/0 excl 10 hours of self study	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	According to the Council of Supply Chain Management Professionals logistics is defined as the process of planning, implementing, and controlling procedures for the efficient and effective transportation and storage of goods including services, and related information from the point of origin to the point of consumption for the purpose of conforming to customer requirements. Logistics activities should be in line with other business activities such as marketing and purchasing, in order to create sustainable competitive advantages for individual companies and service providers, and for the supply chain as a whole. Given the strategic importance of logistics for individual organizations and supply chains, it is vital to know how the performance of logistics activities is measured, and how these activities can be modelled and improved. Therefore in this course different logistics aspects are discussed and analyzed. A selection of quantitative methodologies are introduced to model and provide solutions to a wide range of relevant problems in logistics. We focus on two classes of logistics problems: Logistics problems within organizations: forecasting and demand modeling; inventory management; clustering and classification of items. Logistics problems between organizations: distribution and transportation problems. Methodologies include analytical models, linear programming models, basic data mining techniques.	
Study Goals	At the end of this course you will be able to: Analyze relevant aspects of logistics and transportation system: inventory, routing, location, data; Formulate (and solve) a selection of relevant logistics problems; Select and apply an appropriate methodology for a given logistics problem; Find possibilities to improve/adapt a logistics system through data and analytical models; Model operational aspects through mathematical linear programming models; Understand the integrative role of logistics function in and between organizations;	
Education Method	Lectures Assignments Self-study The educational activities will be held on campus, and attendance is highly recommended to complete the course. Consultation hours will be held both online and on campus.	
Literature and Study Materials	A selection of chapters from: 1) Reid, R. D., Sanders, N. R., Operations Management: An Integrated Approach, 5th edition, Wiley, 2013. 2) Daganzo, C.F., Logistics Systems Analysis, 4th edition, Springer-Verlag, 2010. 3) Zaki, Mohammed J., Wagner Meira Jr, and Wagner Meira. Data mining and analysis: fundamental concepts and algorithms. Cambridge University Press, 2014. 4) A booklet provided at the start of the course 5) Slides and exercises	
Prerequisites	Logistiek 1, Basic mathematics	
Assessment	The course consists of two parts with the following weighting: Logistics problems within organizations: 50%. Distribution problems: 50%. Grading: One written exam: 80%. This is a closed-book exam with the possibility of bringing one formula sheet created by the student on a single A4 paper. The A4 can be written on both sides. 2 Group Assignments (one for each part of the course): 20%. Expected group size: 4 students. The passing grade is 5.8, calculated as: $0.8 * (\text{final exam}) + 0.2 * ((\text{Assignment 1} + \text{Assignment 2})/2)$. Minimum grade for final exam: minimum grade is 5.0 Minimum grade for assignments: average grade at least 5 $\rightarrow (\text{Assignment 1} + \text{Assignment 2})/2 \geq 5.0$ It is possible to repair the assignments, but the maximum grade that can be obtained for each repaired assignment is 6. The repair	

can also be performed individually, by resubmitting every deliverable.

Any sufficient mark for any assessment can be retained for the next academic year.

In case of unforeseen circumstances, the final exam will take place in an online format: a take-home exam with a limited duration. The student will receive the exam via email at the start of the exam session and must upload the answers in Brightspace within the time limit. In case the regular on-campus exam can be held, no online possibility will be offered.

Exam Hours

3 hours

Year	2023/2024
Organization	Technology, Policy and Management
Education	Bachelor Systems Engineering, Policy Analysis & M

Technologiespecialisatie 4	
Director of Studies	A.M.G. Zuiderwijk-van Eijk

TB242EB	Processes in the Energy Sector	5
Module Manager	Dr.ir. L. Stougie	
Instructor	Dr.ir. L. Stougie	
Responsible for assignments	Dr.ir. L. Stougie	
Co-responsible for assignments	Dr.ir. I. Bouwmans	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	

TB242IC	Interconnected Engineering Systems	5
Module Manager	Dr. Y. Ding	
Instructor	R.I.J. Dobbe	
Co-responsible for assignments	Prof.dr.ir. M.F.W.H.A. Janssen	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Internet is a global system that has interconnected various kinds of systems and services worldwide. This course analyzes the architecture and core technologies of the Internet from systemic and interdisciplinary perspectives. As a hands-on oriented course, we learn network programming (using python) and apply the learned programming skills in practical settings. Attention is paid to the increasing importance of interconnected environments, in particular the networking of engineering systems and services through digital platforms. We analyse the concept of interconnected engineering systems on the basis of concrete cases, e.g., connected cars, cyber-physical systems, as means to illustrate the complex interactions of technology and society. In addition, a number of modern development and research themes will be discussed for the evolution of distributed information and communication systems, e.g., new networking and computing techniques for large-scale deployment of the Internet of Things (IoT). On top of the key technologies, we consider the social context to which the high expectations of the future Internet technologies need to tackle in terms of security, privacy risks.	
Study Goals	After the course, students are able to: - Illustrate Internet architecture, design principles and routing algorithms - Evaluate computing paradigms and wireless & mobile communication in practical scenarios - Design networking programming solutions for practical problems - Apply Internet of Things (IoT) and cyber-physical systems (CPS) technologies in case studies - Analyze security & privacy risks, challenges and solutions for interconnected engineering systems such as IoT	
Education Method	Lectures, guest lectures, individual and group work, programming tutorials and presentation practice	
Assessment	- Written Final Exam (Digital): 80% of final grade - Programming Group Project (including mandatory assignments, presentations): 20% of final grade	
	In case of unforeseen circumstances or measures resulting from e.g., COVID-19, the prescribed "Written Digital Final Exam" will be conducted as time-based take-home exam.	

TB242TB	Improving the Transport System	5
Module Manager	Dr. J.A. Annema	
Instructor	Dr. J.A. Annema	
Instructor	N. Mouter	
Responsible for assignments	Dr. J.A. Annema	
Co-responsible for assignments	N. Mouter	
Contact Hours / Week x/x/x/x	0/x/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	

Year	2023/2024
Organization	Technology, Policy and Management
Education	Bachelor Systems Engineering, Policy Analysis & M

Technologiespecialisatie 5	
Director of Studies	A.M.G. Zuiderwijk-van Eijk

CTB1420-17	Transport en Planning	5
Responsible Instructor	Dr.ir. A. Gavriilidou	
Instructor	Dr. V.L. Knoop	
Practical Coordinator	Dr.ir. A. Gavriilidou	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	Dutch	
Collegerama	No	

TB243EA	Analysis of Industrial Systems	5
Course Coordinator	Dr.ir. P. Ibarra Gonzalez	
Instructor	Dr.ir. P. Ibarra Gonzalez	
Responsible for assignments	Prof.dr.ir. Z. Lukszo	
Co-responsible for assignments	Dr.ir. L. Stougie	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	TB141E, TB142E, TB241E	
Summary	The course aims to teach students a number of relevant production processes from the level of unit operations up to the system level. The students will learn how to recognize the basic elements of a production process (flowsheet-graphical way of describing a process) and understand their role and sequence. It also gives an overview of important industries in the Netherlands.	
Course Contents	The main topics covered in this course are: - Structuring industrial production processes - Separation and reaction processes - Wastewater treatment and drinking water preparation - Batch and Continuous Processes - The methanol process as an example of a continuous process - The brewery as an example of a batch process - The crude oil system: Refinery - Petrochemical industry - Overview of Processes in the Agro-Industry - Overview of Processes from the Biobased Industry - Industrial processes in the Netherlands	
Study Goals	By the end of the course, you should be able to: LO1- Identify and name the feedstock, unit operations, and products of selected industrial production processes LO2- Design industrial processes using flow sheeting tools LO3- Describe the processing steps, required unit operations, and product chains of selected industrial production processes LO4- Explain the basic functioning and types of reactors and separation units in selected industrial production processes LO5- Calculate the performance of a simple production process through mass and energy balances LO6- Calculate product conversion for different types of reactors LO7- Determine variables influencing the performance of separation processes	
Education Method	Prior reading of course material (self-study), interactive lectures, instruction lectures (tutorial-style), class assignments, and homework assignments.	
Computer Use	Use of a laptop is needed to answer the assignments in Excel. Please bring it to the Thursday and Friday lectures.	
Literature and Study Materials	If required, before each lecture extra material for reading will be provided.	
Books	A course manual will be provided	
Assessment	Formative assessments during lectures, both individually and in groups (oral participation, exercises in Excel and short multiple-choice tests). The summative assessment is in the form of a final exam in week 10 and has a duration of 3 hours. The exam consists of 2 parts. The first part consists of multiple-choice questions that count for 50% of the final course mark. The second part includes open questions that count for the other 50% of the final course mark.	
Exam Hours	3	

TB243IB	I&C Systems: Risk & Control	5
Module Manager	Dr. Y. Ding	
Instructor	S.H. van Engelenburg	
Instructor	Dr. Ir. M. Tuler de Oliveira	
Co-responsible for assignments	Prof.dr.ir. M.F.W.H.A. Janssen	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This module is about assessing and overcoming risks in ICT to ensure systems are maintainable in the long term in complex multi-actor environments and not create new risks. You learn to understand the different hardware components that make up ICT systems and the different organizational, inter-organizational, and large-scale systems designs. You learn how to use this knowledge to assess the risks different ICT systems pose for individuals, organizations, and society. You will also learn the various strategies to reduce these risks and determine if and how to apply them. The theoretical background will be explained during the first part of the lectures. In the second part, in groups of 3-6 students, you will then apply the theory to several case studies in which you will assess risks and identify strategies for reducing them. You will present your case to the other students in the course, after which they can ask questions about it, and there will be a more in-depth discussion of the case, the strategies chosen, and the considerations made.	
Study Goals	- To describe the components of ICT systems (network devices, servers, workstations). - To describe the different designs for organisational, inter-organisational and large-scale ICT systems. - To describe different hardware and management strategies for reducing risks in ICT. - To assess the risks of an ICT system on the systems level, within organisations and on an inter-organisational and societal level in a given case. - To recommend an appropriate strategy for reducing risks for an ICT system in a given case.	
Education Method	Lectures, assignments, self-study	
Assessment	The assessment during the course consists of two parts. The first one is an individual exam, which contains a mix of knowledge questions and simple cases for which you will need to assess and address risks. The individual exam will be 60% of your final grade. The second part of your assessment will be based on a case study assignment for which you will assess risks and identify a strategy for dealing with those risks in your group during the course. For this part, the assignment report and its presentation are done in groups (3-6 students) and will be 40% of your final grade. Participation in the peer-review feedback sessions is required as a pass/fail on this course. You need to score at least a 5 for each part and at least a 5.75 for both parts together to pass the course	

Year	2023/2024
Organization	Technology, Policy and Management
Education	Bachelor Systems Engineering, Policy Analysis & M

3e jaar BSc TB 2023	
Director of Studies	A.M.G. Zuiderwijk-van Eijk

Year	2023/2024
Organization	Technology, Policy and Management
Education	Bachelor Systems Engineering, Policy Analysis & M

Verplichte vakken 3e jaar 2023	
Director of Studies	A.M.G. Zuiderwijk-van Eijk

TB323A	Governance van sociotechnische systemen	5
Module Manager	Dr. R.S. van Wegberg	
Instructor	Dr. H.G. van der Voort	
Instructor	Dr. M.L.C. de Bruijne	
Instructor	Dr.ir. E. Minkman	
Instructor	Dr. R.S. van Wegberg	
Co-responsible for assignments	Dr. M.L.C. de Bruijne	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	Dutch	

TB351D	Bachelor Final Project	15
Module Manager	M. Kroesen	
Co-responsible for assignments	Dr. P.W.G. Bots	
Contact Hours / Week x/x/x/x	2/2/2/2	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	Dutch	

Year	2023/2024
Organization	Technology, Policy and Management
Education	Bachelor Systems Engineering, Policy Analysis & M

Governance specialisatie (E&I / I&C / T&L)	
Director of Studies	A.M.G. Zuiderwijk-van Eijk

TB321EB	Governance specialisation (E and I)	5
Module Manager	Prof.dr.ir. L.J. de Vries	
Instructor	M. Looij	
Co-responsible for assignments	Dr. A.F. Correlje	
Contact Hours / Week x/x/x/x	0/0/x/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	

TB321IB	Governance specialisation (I and C)	5
Module Manager	A.M.G. Zuiderwijk-van Eijk	
Instructor	M. Looij	
Co-responsible for assignments	Dr. J. Ubacht	
Contact Hours / Week x/x/x/x	0/0/x/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	

TB321TB	Governance specialisation (T and L)	5
Module Manager	Dr. J.A. Annema	
Instructor	M. Looij	
Co-responsible for assignments	N. Mouder	
Contact Hours / Week x/x/x/x	0/0/x/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	

Year	2023/2024
Organization	Technology, Policy and Management
Education	Bachelor Systems Engineering, Policy Analysis & M

Technologiespecialisatie 6	
Director of Studies	A.M.G. Zuiderwijk-van Eijk

TB341EC	Performance analysis in Energy and Industry	5
Module Manager	Dr.ir. P.W. Heijnen	
Instructor	Dr. P.W.G. Bots	
Co-responsible for assignments	Dr. P.W.G. Bots	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3A 3B 4A	
Course Language	Dutch	

TB341ID	I&C in a socio-technical context	5
Module Manager	Dr.ir. G.A. de Reuver	
Instructor	Dr. J. Ubacht	
Instructor	R.I.J. Dobbe	
Responsible for assignments	Dr.ir. G.A. de Reuver	
Co-responsible for assignments	R.I.J. Dobbe	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	

TB341TC	Quantitative Models for Transport	5
Module Manager	Dr. S. Fazi	
Instructor	B. Pudane	
Instructor	Dr. S. Balakrishnan	
Co-responsible for assignments	Dr. J.A. Annema	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4A	
Course Language	English	
Course Contents	Many transportation problems are tackled with quantitative methods, due to the necessity for the decision-makers to quantify relevant aspects: service frequencies, sequence of visits, costs, timings, etc. In this course, we will propose three fundamental methodologies in the transport domain to give answers to problems related to passengers mobility and freight transport. In the first part of the course, Discrete choice modelling will be proposed to solve passengers mobility related problems. The goal is to understand how individuals choose between different transport options based on a set of features and related utility functions. In the second part, simulation and mixed-integer linear programming models will be proposed to model both operational and strategic decisions. The proposed models are extensively used in container transport, parcel delivery, and trucking problems. Commercial optimization and simulation software will be used to solve the models and analyse the solutions.	
Study Goals	At the end of this course you will be able to: - Apply basic discrete choice modelling techniques - Perform statistical analysis to draw the main elements of the choice models - Model transportation systems by means of conceptual models and simulation - Recognize and develop the right mathematical model for a given transportation problem; - Analyse solutions and perform sensitivity analyses with ad-hoc software.	
Education Method	Lectures Assignments Self-study	
Assessment	Consultation hours will be held both online and on-campus. The course consists of two parts with the following weighting: Discrete choice models: 50%. Combinatorial linear mathematical models and simulation: 50%. Grading: One written exam: 80%. This is a closed-book exam with the possibility of bringing one formula sheet created by the student on a single A4 paper. The A4 can be written on both sides. 2 Group Assignments (one for each part of the course): 20%. Expected group size: 4 students. The passing grade is 5.8, calculated as: $0.8 * (\text{final exam}) + 0.2 * ((\text{Assignment 1} + \text{Assignment 2})/2)$. Minimum grade for the final exam: minimum grade is 5.0 Minimum grade for assignments: average grade at least 5 $\rightarrow (\text{Assignment 1} + \text{Assignment 2})/2 \geq 5.0$ It is possible to repair the assignments, but the maximum grade that can be obtained for each repaired assignment is 6. The repair can also be performed individually, by resubmitting every deliverable. Any sufficient mark for any assessment can be retained for the next academic year. In case of unforeseen circumstances, the final exam will take place in an online format: a take-home exam with a limited duration. The student will receive the exam via email at the start of the exam session and must upload the answers in Brightspace within the time limit. In case the regular on-campus exam can be held, no online possibility will be offered.	

Dr. J.A. Annema

Unit	Techniek, Bestuur & Management
Department	Transport en Logistiek
Telephone	+31 15 27 88912
Room	31.a3.230

Dr.ir. W.L. Auping

Unit	Techniek, Bestuur & Management
Department	Beleidsanalyse
Telephone	+31 15 27 89734
Room	31.B2.290

S. Azimi Rashti

Department	Informatie en Communicatie Technologie
Telephone	+31 15 27 86364

Dr. S. Balakrishnan

Department	Transport en Logistiek
Telephone	+31 15 27 85831

Dr.ir. G. Bekebrede

Unit	Techniek, Bestuur & Management
Department	Organisatie en Governance
Telephone	+31 15 27 81139
Room	31.b2.060

Dr.ir. R. van Bergem

Department	Economie van Technologie en Innovatie
Telephone	+31 15 27 84819

Dr. P.W.G. Bots

Unit	Techniek, Bestuur & Management
Department	Beleidsanalyse
Telephone	+31 15 27 84538
Room	31.b1.330

Dr.ir. I. Bouwmans

Unit	Techniek, Bestuur & Management
Department	Energie and Industrie
Telephone	+31 15 27 84807
Room	31.a3.240

Prof.mr.dr. J.A. de Bruijn

Unit	Techniek, Bestuur & Management
Department	Organisatie en Governance
Telephone	+31 15 27 87169

Dr. M.L.C. de Bruijne

Unit	Techniek, Bestuur & Management
Department	Organisatie en Governance
Telephone	+31 15 27 82323
Room	31.B2.080

Dr.ir. E.J.L. Chappin

Unit	Techniek, Bestuur & Management
Department	Energie and Industrie
Telephone	+31 15 27 83410
Room	31.A3.280

Dr. S.M. Copeland

Unit	Techniek, Bestuur & Management
Department	Ethiek & Filosofie van Techniek

Telephone	+31 15 27 87723
------------------	-----------------

Dr. A.F. Correlje

Unit	Techniek, Bestuur & Management
Department	Economie van Technologie en Innovatie
Telephone	+31 15 27 87924

Dr.ir. V.J. Cortes Arevalo

Department	Beleidsanalyse
Telephone	+31 15 27 87861

Ir. S. van Cranenburgh

Unit	Techniek, Bestuur & Management
Department	Transport en Logistiek
Telephone	+31 15 27 86957
Room	31.A3.210

F.M. d'Hont

Department	Beleidsanalyse
Telephone	+31 15 27 88332

Dr.ir. J. De Stefani

Department	Informatie en Communicatie Technologie
Telephone	+31 15 27 83829

Dr. Y. Ding

Unit	Techniek, Bestuur & Management
Department	Informatie en Communicatie Technologie
Telephone	+31 15 27 85659
Room	31.B3.260

R.I.J. Dobbe

Department	Informatie en Communicatie Technologie
Telephone	+31 15 27 89742

Dr. J.H.R. van Duin

Unit	Techniek, Bestuur & Management
Department	Transport en Logistiek
Telephone	+31 15 27 81142
Room	31.a3.210

I. Dursun

Department	Transport en Logistiek
Telephone	+31 15 27 84965

S.H. van Engelenburg

Department	Informatie en Communicatie Technologie
Telephone	+31 15 27 82972

Dr. S. Fazi

Department	Transport en Logistiek
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Dr.ir. A. Gavrilidou

Department	Transport and Planning
Room	23.HG 4.40

Prof.dr.ir. P.H.A.J.M. van Gelder

Unit	Techniek, Bestuur & Management
Department	Safety and Security Science

Telephone	+31 15 27 86544
Room	31.c1.170

Dr.ir. P.W. Heijnen

Unit	Techniek, Bestuur & Management
Department	Energie and Industrie
Telephone	+31 15 27 88072
Room	31.a3.240

Dr.ir. P. Ibarra Gonzalez

Department	Energie and Industrie
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Department	Energie and Industrie
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Prof.dr.ir. M.F.W.H.A. Janssen

Unit	Techniek, Bestuur & Management
Department	Engineering, Systems and Services
Telephone	+31 15 27 81140
Room	31.b3.150

Dr. O. Kammouh

Department	Systeemkunde
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Dr. V.L. Knoop

Unit	Civiele Techniek & Geowetensch
Department	Transport and Planning
Telephone	+31 15 27 88413
Room	23.HG 4.37

mr. J.M. Kooijman

Unit	Techniek, Bestuur & Management
Department	Organisatie en Governance
Telephone	+31 15 27 81173
Room	31.b2.220

Dr.ir. G. Korevaar

Unit	Techniek, Bestuur & Management
Department	Energie and Industrie
Telephone	+31 15 27 83659
Room	31.A3.200

Dr. L.J. Kortmann

Unit	Techniek, Bestuur & Management
Department	Beleidsanalyse
Telephone	+31 15 27 82154

M. Kroesen

Unit	Techniek, Bestuur & Management
Department	Transport en Logistiek
Telephone	+31 15 27 87183
Room	31.A3.250

Dr. M.D. - Kubli

Department	Beleidsanalyse
-------------------	----------------

N. van der Linden

Department	Beleidsanalyse
Room	31.B2.330

M. Looij

Department	Instituut voor Talen en Academische Vaardigheden
-------------------	--

Telephone Room	+31 15 27 89821 31.C0.080
---------------------------	------------------------------

Prof.dr.ir. Z. Lukszo

Unit Department	Techniek, Bestuur & Management Energie and Industrie
Telephone Room	+31 15 27 81147 31.B3.030

Dr. N. Metoui

Department	Informatie en Communicatie Technologie
Telephone	+31 15 27 84865

M.B. van der Meulen

M.B. van der Meulen

Department	Sports & Games
Telephone Room	+31 15 27 82399 31.C0.080

Unit Department	Techniek, Bestuur & Management Instituut voor Talen en Academische Vaardigheden
Telephone Room	+31 15 27 82399 31.C0.080

Dr.ir. E. Minkman

Department	Organisatie en Governance
-------------------	---------------------------

Dr. E.J.E. Molin

Unit Department	Techniek, Bestuur & Management Transport en Logistiek
Telephone Room	+31 15 27 88510 31.a3.250

N. Mouter

Unit Department	Techniek, Bestuur & Management Transport en Logistiek
Telephone Room	+31 15 27 89748 31.A3.060

Dr.ir. I. Nikolic

Unit Department	Techniek, Bestuur & Management Systeemkunde
Telephone Room	+31 15 27 81135 31.B1.050

K.L.L. van Nunen

Department	Safety and Security Science
Telephone Room	+31 15 27 83264 31.C1.060

Ö. Okur

Department	Systeemkunde
-------------------	--------------

B. Pudane

Department	Transport en Logistiek
Telephone	+31 15 27 86903

S. Renes

Department	Economie van Technologie en Innovatie
Telephone Room	+31 15 27 87566 31.C3.090

S. Renes

Department	Services
-------------------	----------

Dr.ir. G.A. de Reuver

Unit	Techniek, Bestuur & Management
Department	Informatie en Communicatie Technologie
Telephone	+31 15 27 81920
Room	31.B3.210

J. Rezaei

Unit	Techniek, Bestuur & Management
Department	Transport en Logistiek
Telephone	+31 15 27 81716
Room	31.B3.100

Dr.ir. M.P.M. Ruijgh-van der Ploeg

Unit	Techniek, Bestuur & Management
Department	Beleidsanalyse
Telephone	+31 15 27 81867
Room	31.b1.350

Dr.ir. L. Stougie

Unit	Techniek, Bestuur & Management
Department	Energie and Industrie
Telephone	+31 15 27 87041
Room	31.a3.320

Dr.ir. H. Torkamaan

Department	Systeemkunde
Telephone	+31 15 27 86428
Room	31.B1.270

Dr. Ir. M. Tuler de Oliveira

Department	Informatie en Communicatie Technologie
-------------------	--

Dr. J. Ubacht

Unit	Techniek, Bestuur & Management
Department	Informatie en Communicatie Technologie
Telephone	+31 15 27 82791
Room	31.b3.190

Drs. J.A. Verheij

Unit	Elektrotechn., Wisk. & Inform.
Department	Analysis
Telephone	+31 15 27 85044
Room	28.1.W740

Dr. H.G. van der Voort

Unit	Techniek, Bestuur & Management
Department	Organisatie en Governance
Telephone	+31 15 27 88541
Room	31.b2.140

Prof.dr.ir. L.J. de Vries

Unit	Techniek, Bestuur & Management
Department	Energie and Industrie
Telephone	+31 15 27 81137
Room	31.B3.030

Dr. T.W.C. Vroegrijk

Unit	Elektrotechn., Wisk. & Inform.
Department	Analysis
Telephone	+31 15 27 88267

Room	36.HB 08.260
-------------	--------------

Prof.dr. M.E. Warnier

Unit	Techniek, Bestuur & Management
Department	Multi Actor Systems
Telephone	+31 15 27 82232

Dr. R.S. van Wegberg

Department	Organisatie en Governance
Telephone	+31 15 27 81320
Room	31.B2.210

Prof.dr.ir. M.P.C. Weijnen

Unit	Techniek, Bestuur & Management
Department	Energie and Industrie
Telephone	+31 15 27 88074
Room	31.b3.010

A.M.G. Zuiderwijk-van Eijk

Unit	Techniek, Bestuur & Management
Department	Informatie en Communicatie Technologie
Telephone	+31 15 27 85828
Room	31.B3.220
